

CAMPBELL AAF, KY, USA

WMO: 746710

Lat: 36.667N

Lon: 87.483W

Elev: 175

StdP: 99.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-8.6	-17.0	0.9	-9.3	-14.0	1.1	-6.7	10.9	10.3	9.6	7.4	2.9	350	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.5	24.4	33.0	24.3	32.2	24.0	26.3	31.7	25.7	30.8	25.2	30.0	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.5	28.5	24.2	19.6	28.0	23.8	19.1	27.7	82.6	31.8	80.3	31.0	77.8	29.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.2	8.1	7.2		-15.6	36.1	4.0	2.2	-18.5	37.7	-20.8	38.9	-23.1	40.2	-26.0	41.7	
(5)			DB	-15.6	27.7	4.3	0.8	-18.7	28.3	-21.2	28.7	-23.5	29.2	-26.6	29.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	2.5	5.2	9.7	15.2	20.2	24.4	26.0	25.5	22.0	15.8	9.2	4.8
	DBStd	9.57	6.87	6.53	6.00	5.09	4.05	2.83	2.42	2.53	3.77	4.76	5.31	5.88
	HDD10.0	766	247	159	80	13	0	0	0	0	0	7	77	182
	HDD18.3	2094	490	368	272	118	31	1	0	0	12	107	275	420
	CDD10.0	2627	16	26	71	171	316	432	496	481	359	187	53	20
	CDD18.3	916	0	1	6	25	88	183	237	223	121	29	2	1
	CDH23.3	8753	0	3	51	216	730	1741	2402	2208	1122	265	14	1
	CDH26.7	3450	0	0	7	36	205	693	1044	965	438	62	1	0
(14) Wind	WSAvg	3.1	3.6	3.7	3.6	3.7	3.0	2.7	2.5	2.3	2.4	2.7	3.1	3.3
(15) Precipitation	PrecAvg	1239	102	99	115	112	132	101	108	74	83	94	98	122
	PrecMax	1480	240	217	331	204	291	301	305	244	246	226	237	321
	PrecMin	938	35	35	48	43	41	20	26	8	12	0	21	36
	PrecStd	158	42	47	59	47	57	56	62	47	55	56	55	57
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	22.9	27.5	29.4	32.4	34.6	36.3	37.2	35.1	31.0	25.2	22.0
		MCWB	16.8	16.3	17.5	19.7	22.8	24.8	25.4	24.3	22.3	21.6	17.8	17.7
	2%	DB	18.6	20.4	24.6	27.7	30.2	32.9	34.1	34.6	32.8	28.3	22.5	18.8
		MCWB	16.0	15.6	16.4	18.8	22.1	24.3	25.3	24.2	22.5	20.2	16.1	15.8
	5%	DB	16.2	18.0	22.2	26.0	29.0	31.9	32.9	32.9	31.3	26.3	20.2	16.4
		MCWB	13.6	13.6	15.1	17.8	21.4	24.1	25.1	24.1	22.3	18.7	15.4	13.7
	10%	DB	13.4	15.7	19.7	24.0	27.6	30.7	31.9	31.6	29.3	24.1	17.9	14.1
		MCWB	10.3	11.9	13.8	17.1	20.9	23.6	24.7	23.7	21.6	18.0	14.1	11.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.4	19.1	21.9	24.7	26.6	27.4	26.9	25.3	23.4	19.7	18.5
		MCDB	19.3	20.5	24.3	26.9	29.5	32.0	32.8	32.5	30.2	27.4	22.4	20.6
	2%	WB	16.3	16.9	17.8	20.6	23.6	25.7	26.6	26.0	24.3	22.1	18.1	16.6
		MCDB	17.9	19.7	22.5	25.3	28.3	31.0	32.0	31.3	29.0	26.1	20.8	18.4
	5%	WB	14.2	14.9	16.5	19.5	22.7	25.1	25.9	25.4	23.5	20.8	16.5	14.4
		MCDB	16.2	17.2	20.5	23.8	27.2	30.1	31.1	30.3	28.2	24.0	19.2	16.0
	10%	WB	11.0	12.5	14.9	18.3	21.9	24.4	25.4	24.8	22.8	19.4	14.7	12.0
		MCDB	12.9	15.1	19.0	22.3	25.9	29.1	30.3	29.3	27.3	23.0	17.4	13.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.2	11.3	11.9	10.6	10.4	10.3	11.0	12.0	12.2	11.0	9.1
		MCDBR	11.3	13.1	14.3	13.3	11.9	11.5	11.5	12.6	13.8	14.1	13.5	11.5
	5% WB	MCWBR	8.9	9.1	8.1	6.9	5.3	4.7	4.4	4.6	5.0	6.9	8.5	8.8
		MCDWR	10.3	11.1	12.3	10.9	10.1	10.2	10.2	10.4	10.5	11.0	11.0	10.0
(40) Clear-Sky Solar Irradiance	taub		0.315	0.323	0.371	0.400	0.439	0.458	0.495	0.496	0.421	0.366	0.329	0.309
		taud	2.480	2.439	2.348	2.269	2.208	2.180	2.098	2.083	2.284	2.426	2.481	2.505
	Ebn at Noon		878	914	893	882	849	830	795	785	830	853	853	861
		Edh at Noon	83	97	116	133	144	148	159	158	123	97	82	76
(44) All-Sky Solar Radiation	RadAvg		2.17	2.75	3.83	5.01	5.69	6.33	6.06	5.56	4.68	3.58	2.52	1.84
	RadStd		0.19	0.35	0.31	0.32	0.49	0.43	0.35	0.27	0.33	0.40	0.26	0.18

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+71

Nomenclature: See separate page